

Start the class Docker container

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Prerequisites

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1. **Docker Desktop** installed on your computer.

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 - Download it from Docker Desktop.

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 - Download it from Docker Desktop.
 - Follow the installation instructions for your operating system.

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1. **Docker Desktop** installed on your computer.
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Open Docker Desktop and navigate to the “Images” tab:



Containers



Images



Volumes



Builds



Docker Scout



Extensions

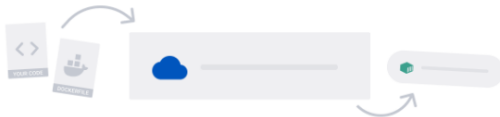


Images

[Give feedback](#)

Local

Hub repositories



Images are used to run containers

You can either build an image from a Dockerfile, or download an existing image to run

[Search images to run](#)docker
hub-image

Run Docker Hub images

5 mins



Push to Hub

Publish your image

5 mins



Engine running



RAM 1.01 GB CPU 0.12% Disk 62.37 GB avail. of 67.32 GB



New version available

2. The Dockerfile provided for the class.

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Save this in a file named **Dockerfile**:

Prerequisites

2. The Dockerfile provided for the class.

Save this in a file named **Dockerfile**:

```
FROM ubuntu:24.04
ENV DEBIAN_FRONTEND=noninteractive
RUN apt-get update && apt-get install -y --no-install-recommends \
    python3 python3-pip \
    bash openssh-server
RUN mkdir /var/run/sshd && \
    echo 'root:rootpassword' | chpasswd
EXPOSE 22
CMD ["/usr/sbin/sshd", "-D"]
```

Step 1: Build the Docker image

1. Open a terminal:

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- Navigate to the directory where your **Dockerfile** is saved.

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2. Run the build command:
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```
docker build -t class-docker .
```

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1. Open a terminal:

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```
docker build -t class-docker .
```

3. Verify the image:

Step 1: Build the Docker image

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- Navigate to the directory where your **Dockerfile** is saved.

2. Run the build command:

- Use the terminal to build the image:

```
docker build -t class-docker .
```

3. Verify the image:

- In the “Images” tab, check that **class-docker** is listed.

 Containers

 **Images**

 Volumes

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 Docker Scout

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Images

[Give feedback](#)

Local

Hub repositories

0 Bytes / 211.99 MB in use 1 images

Last refresh: 3 hours ago 

Search



Name

Tag

Image ID

Created

Size

Actions



class-dor latest

a0c5fe6247b4

1 day ago

296.58 MB



Showing 1 item

Walkthroughs



 Engine running



RAM 1.23 GB CPU 0.12% Disk 62.49 GB avail. of 67.32 GB

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Step 2: Run the Docker Container

1. Launch the container from Docker Desktop:

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
 - Go to the “Images” tab.

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
 - Go to the “Images” tab.
 - Click the play icon next to the `class-docker` image.

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
 - Go to the “Images” tab.
 - Click the play icon next to the `class-docker` image.
 - Set the following options:

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
 - Go to the “Images” tab.
 - Click the play icon next to the `class-docker` image.
 - Set the following options:
 - Name: `class-container`

Step 2: Run the Docker Container

1. Launch the container from Docker Desktop:

- Go to the “Images” tab.
- Click the play icon next to the **class-docker** image.
- Set the following options:
 - **Name:** **class-container**
 - **Ports:** Map “Container port” 22 to “Host port” 2222.

Step 2: Run the Docker Container

1. Launch the container from Docker Desktop:

- Go to the “Images” tab.
- Click the play icon next to the **class-docker** image.
- Set the following options:
 - **Name:** **class-container**
 - **Ports:** Map “Container port” 22 to “Host port” 2222.
 - **Volumes:** In “Host path,” enter the directory that you want to share with the container. In “Container path,” enter **/workspace**

Step 2: Run the Docker Container

1. Launch the container from Docker Desktop:

- Go to the “Images” tab.
- Click the play icon next to the **class-docker** image.
- Set the following options:
 - **Name:** **class-container**
 - **Ports:** Map “Container port” 22 to “Host port” 2222.
 - **Volumes:** In “Host path,” enter the directory that you want to share with the container. In “Container path,” enter **/workspace**

Step 2: Run the Docker Container

1. Launch the container from Docker Desktop:

- Go to the “Images” tab.
- Click the play icon next to the **class-docker** image.
- Set the following options:
 - **Name:** **class-container**
 - **Ports:** Map “Container port” 22 to “Host port” 2222.
 - **Volumes:** In “Host path,” enter the directory that you want to share with the container. In “Container path,” enter **/workspace**

Setting a Volume is important because only files stored in that directory will survive if the container is destroyed!

Step 2: Run the Docker Container

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```
docker run -dit \  
    --name class-container \  
    --publish 2222:22 \  
    --volume /home/user/cogs205b:/workspace \  
    class-docker
```

Step 2: Run the Docker Container

2. Alternatively, you can use the terminal to run the container:

```
docker run -dit \  
    --name class-container \  
    --publish 2222:22 \  
    --volume /home/user/cogs205b:/workspace \  
    class-docker
```

(But don't do this now, use Docker Desktop.)

Step 2: Run the Docker Container

3. Verify the container is running:

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- In Docker Desktop, check that the `class-container` status shows as “Running.”

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3. Verify the container is running:

- In Docker Desktop, check that the `class-container` status shows as “Running.”
- Or, use the terminal:

```
docker ps
```


 Containers

 Images

 Volumes

 Builds

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[Containers](#) / class-container

class-container



363730a5e735 
2222:22 



[class-docker:latest](#)

STATUS

Running (0 seconds ago)



Logs

Inspect

Bind mounts

Exec

Files

Stats



 Engine running

|| ⋮

RAM 1.14 GB CPU 0.00% Disk 62.37 GB avail. of 67.32 GB

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Step 3: Test the container

Access the container

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1. Use Docker Desktop's integrated shell, and test the Python installation:

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Access the container

1. Use Docker Desktop's integrated shell, and test the Python installation:

```
python3 --version
```


Containers

Images

Volumes

Builds

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Extensions



[Containers](#) / class-container

class-container



363730a5e735



[class-docker:latest](#)

[2222:22](#)

STATUS

Running (2 minutes ago)



Logs

Inspect

Bind mounts

Exec

Files

Stats



Debug mode

[Open in external terminal](#)

```
# python3 --version
Python 3.12.3
#
```



 Engine running



RAM 1.16 GB CPU 0.00% Disk 62.37 GB avail. of 67.32 GB

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Step 3: Test the container

The integrated shell is running on the Ubuntu/Linux operating system inside the container. You can learn the basic commands of the command line interface from the cheat sheet I'll put on GitHub.

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Alternatively, open your system's terminal to connect to the container:

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The integrated shell is running on the Ubuntu/Linux operating system inside the container. You can learn the basic commands of the command line interface from the cheat sheet I'll put on GitHub.

Alternatively, open your system's terminal to connect to the container:

```
docker exec -it class-container bash
```

When you're done: Stop (and later restart) the container

In Docker Desktop, click the stop or play button in the Containers tab.

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Or, in the terminal to stop the container:

```
docker stop class-container
```


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```
docker stop class-container
```

In the terminal to restart the container:

When you're done: Stop (and later restart) the container

In Docker Desktop, click the stop or play button in the Containers tab.

Or, in the terminal to stop the container:

```
docker stop class-container
```

In the terminal to restart the container:

```
docker start class-container
```

You've now set up the class Docker container, confirmed it works, and learned how to manage it using Docker Desktop. This container will be your consistent Python environment for all class activities.

You've now set up the class Docker container, confirmed it works, and learned how to manage it using Docker Desktop. This container will be your consistent Python environment for all class activities.

In the future, you could make new containers for new projects and manage them in Docker Desktop or submit them to Docker Hub.